

Khush A. Patel

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Objective

Mechanical Engineering student at Johns Hopkins University with research and industry experience in computational engineering, robotics, and electromechanical product development seeking a Summer 2027 mechanical engineering internship.

Education

Johns Hopkins University, Whiting School of Engineering

Baltimore, MD

B.S. Mechanical Engineering **GPA: 4.00/4.00**

Anticipated May 2028

Relevant Coursework: Statics, Mechanics, Thermodynamics, Differential Equations, Linear Algebra, Multivariable Calculus, Electricity & Magnetism, Digital Electronics, CAD (SolidWorks, Creo), Python.

Technical Skills

- **CAD & Engineering Analysis:** SolidWorks, PTC Creo, Windchill, ANSYS Mechanical, Fluent, CFX, OpenFOAM, Linux
- **Programming:** Python (NumPy, SciPy), MATLAB, C++, MathCAD, Git
- **Robotics & Controls:** ROS 2, Kinematics, Motion Planning, Trajectory Control, PID, LQR, Model Predictive Control (MPC)
- **Mechanical Design:** GD&T, RSS & Worst-Case Tolerance Stack-Up Analysis, DFM/DFA, DFMEA, Technical Drawings
- **Manufacturing:** CNC Milling, CNC Turning, Wire EDM, Sheet Metal Fabrication, FDM & Resin 3D Printing, Laser Cutting

Engineering Experience

Amphenol

Valley Green, PA

Development Engineering Intern – High Speed I/O Connectors (Current)

May 2026–Aug 2026

- Designed **10+** high-speed connector components in **SolidWorks** for next-generation AI and hyperscale data-center applications.
- Performed engineering validation using hand calculations and **ANSYS Mechanical** to optimize structural performance and material selection.
- Conducted **GD&T** and statistical tolerance stack-up analyses on **5+ production drawings**, selecting **ISO hole/shaft fit classes** and evaluating **MMC/LMC** conditions using **RSS** and **worst-case** methodologies to ensure manufacturability and assembly reliability.
- Collaborated with cross-functional engineering teams during design reviews while applying **DFM** and **DFMEA** principles throughout product development.

Johns Hopkins University – AIRO Laboratory

Baltimore, MD + Remote

Research Assistant – Computational Aerodynamics & Controls (Current)

Dec 2025–Present

- Developed and validated high-fidelity **OpenFOAM** simulations to model the aerodynamic wake of naval vessels.
- Built Python workflows to reduce over **120 GB** of CFD flow-field data into **200 MB reduced-order models (ROMs)** for control-oriented applications.
- Supported autonomous Unmanned Aerial Vehicle (UAV) landing research by integrating computational aerodynamics with **Model Predictive Control (MPC)** algorithms.

Amarkosha Technologies (Start-Up)

Baltimore, MD

Mechanical Lead – Product Development

Jul 2025–Dec 2025

- Led mechanical development of a vortex-induced vibration energy harvester for low-speed wind environments.
- Designed the electromagnetic coil–magnet architecture for mechanical-to-electrical energy conversion.
- Performed transient CFD, fluid-structure interaction (FSI), and modal analyses using **ANSYS Fluent, CFX, and Mechanical**.
- Selected for the Johns Hopkins SPARK Accelerator; project named a BII Grant finalist.

Johns Hopkins Mars Rover Team

Baltimore, MD

Mechanical & Controls Engineer – Manipulator

Sep 2025–Present

- Designed a compact **3-DOF robotic wrist**; performed kinematic and joint torque analyses to select optimal BLDC actuators.
- Designed, tested, and iterated motor mounts, encoder integration, and a cycloidal gearbox assembly to improve manufacturability and drivetrain performance.

Projects

4-DOF Robotic Arm and In Depth ANSYS Static Study

- Designed and built a 4-DOF robotic arm featuring Arduino-based actuation and encoder feedback for closed-loop position control. Mechanics were validated with ANSYS FEA and hand calculations.